

Miss Mehwish: 03184148783

MTH401 (Live Quiz)

MID TERM

PAID VU LMS HANDLING by Mam Mehwish

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1. The annihilator operator for the function, g(x) = 2e^x, is

- D+5
- D+1
- ✓ D
- D^2

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2. Consider the non-homogeneous linear differential equation, any(n)+an−1y(n−1)+⋯+a1y′+a0y=g(x)a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x). If the input function, g(x)=x2e5xg(x) = x^2 e^{5x}, then the assumed particular solution ypy_p could be

- (Ax + B)e^{5x}
- ✓ $(Ax^2 + Bx + C)e^{5x}$
- Ae^{5x} cos 5x + Be^{5x} sin 5x
- Ae^{5x}

---

3. If y=c1eαx+c2e4xy = c_1 e^{\alpha x} + c_2 e^{4x} is the complementary solution of

- ✓ y=c1eαx+c2e4xy = c_1 e^{\alpha x} + c_2 e^{4x}
- y=c1e(α+iβ)x+c2e(α−iβ)xy = c_1 e^{(\alpha+i\beta)x} + c_2 e^{(\alpha-i\beta)x}
- y=c1eαx+c2e−αxy = c_1 e^{\alpha x} + c_2 e^{-\alpha x}
- None of these

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4. If the auxiliary equation am2+bm+c=0a m^2 + b m + c = 0 has the complex roots, m1=α+iβm_1 = \alpha + i\beta and m2=α−iβm_2 = \alpha - i\beta, where α\alpha and β>0\beta > 0, then the solution of the differential equation, ay''+by'+cy=0ay'' + by' + cy = 0, could be

- y=c1e(α+iβ)xy = c_1 e^{(\alpha+i\beta)x}
- y=c1e(α+iβ)x−c2e(α−iβ)xy = c_1 e^{(\alpha+i\beta)x} - c_2 e^{(\alpha-i\beta)x}
- ✓ y=c1e(α+iβ)x+c2e(α−iβ)xy = c_1 e^{(\alpha+i\beta)x} + c_2 e^{(\alpha-i\beta)x}
- y=c2e(α−iβ)xy = c_2 e^{(\alpha-i\beta)x}

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5. If y=2+xy = 2 + x, then which of the following is true for it?

- Its annihilator operator is D
- Its annihilator operator is D^3
- ✓ Its annihilator operator is D^2
- Its annihilator operator is D + 1

---

6. Consider the non-homogeneous linear differential equation any(n)+an−1y(n−1)+⋯+a1y'+a0y=g(x)a_n y^{(n)} + a_{n-1} y^{(n-1)} + \dots + a_1 y' + a_0 y = g(x). If the input function, g(x)=sin⁡(4x)g(x) = \sin(4x), then the assumed particular solution ypy_p could be

- Ae4xA e^{4x}
- (A+B)(cos⁡4x+sin⁡4x)(A + B)(\cos 4x + \sin 4x)
- Ax2+Bx+CA x^2 + B x + C
- ✓ Acos⁡4x+Bsin⁡4xA \cos 4x + B \sin 4x

---

7. An annihilator operator LL for the function f(x)=e3xf(x) = e^{3x}, is

- ✓ D−3D - 3
- 3e3x3 e^{3x}
- e3xe^{3x}
- D−e3xD - e^{3x}

---

8. The polynomial linear differential operators with constant coefficients

- ✓ distributive

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- identical
- inverse
- commute

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9. The solution of the auxiliary equation determines the

- order
- ✓ solutions
- linearity
- degree

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10. For non-homogeneous differential equations, the form of the particular solution, yp=yp1+yp2+⋯+ypny_p = y_{p1} + y_{p2} + \dots + y_{pn}, is a linear combination of all the linearly functions generated by repeated differentiation of the input function g(x)g(x).

- ✓ independent
- dependent

1. For the non-exact differential equation M(x,y)dx+N(x,y)dy=0,M(x, y)dx + N(x, y)dy = 0,M(x,y)dx+N(x,y)dy=0, if ∂N∂x−∂M∂y\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} ∂x∂N −∂y∂M is a function of yyy, then the integrating factor is a:

- ✗ multi-variable function of both x and y
- ✓ function of y
- ✗ function of x
- ✗ function of dx

---

2. The differential equation: dydx+5(ddx)3y−3y=esin⁡x\frac{dy}{dx} + 5 \left(\frac{d}{dx} \right)^3 y - 3y = e^{\sin x} dxdy+5(dxd)3y−3y=esinx is an example of:

- ✗ linear
- ✗ Bernoulli
- ✓ non-exact
- ✗ exact

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3. The differential equation: dydx+y=xy3\frac{dy}{dx} + y = xy^3dxdy+y=xy3 is an example of _____ differential equation.

- ✕ linear
- ✕ Bernoulli
- ✓ non-exact
- ✕ exact

---

4. If the initial amount of a radioactive isotope is 100g. What will be the amount at the end of 30 days such that K=0.043K = 0.043K=0.043?

- ✕ 363.279
- ✕ 380.560
- ✕ 360.351
- ✓ 363.279

---

5. The order of the differential equation: d2ydx2+5(dydx)3−3y=esin⁡x\frac{d^2y}{dx^2} + 5 \left(\frac{dy}{dx} \right)^3 - 3y = e^{\sin x}dx2d2y+5(dxdy)3−3y=esinx is:

- ✕ 0
- ✓ 2
- ✕ 3
- ✕ 1

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6. For f(x,y)=x2y−xy2f(x, y) = x^2y - xy^2f(x,y)=x2y−xy2, f(tx,ty)=…f(tx, ty) = \dotsf(tx,ty)=…

- ✕ t3f(x,y)t^3 f(x, y)t3f(x,y)
- ✕ f(x,y)f(x, y)f(x,y)
- ✓ t2f(x,y)t^2 f(x, y)t2f(x,y)
- ✕ tf(x,y)tf(x, y)tf(x,y)

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7. If initial amount of a radioactive isotope is 100g. What will be the amount at the end of 30 days such that K=0.043K = 0.043K=0.043?

- ✓ 363.279
- ✗ 380.560
- ✗ 360.351

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8. If the differential equation M(x,y)dx+N(x,y)dy=0M(x,y)dx + N(x,y)dy = 0M(x,y)dx+N(x,y)dy=0 is not exact, then after _____ an appropriate function u(x,y)u(x, y)u(x,y), we can form it to be exact.

- ✗ subtracting
- ✓ multiplying by
- ✗ dividing by
- ✗ adding

---

9. Which of the following is an example of partial differential equation?

- ✗ ∂uðy=∂vðx\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}ðyðu=ðxðv
- ✓ ϕ(x)=f(x)+∫abκ(x,t)ϕ(t)dt\varphi(x) = f(x) + \int_a^b \kappa(x, t)\varphi(t)dtϕ(x)=f(x)+∫ab κ(x,t)ϕ(t)dt
- ✗ ax2+bx+c=0,a≠0ax^2 + bx + c = 0, a \neq 0ax2+bx+c=0,a□=0
- ✗ (y−x)dx+4xdy=0(y - x) dx + 4xdy = 0(y−x)dx+4xdy=0

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10. A set of functions {f1(x),f2(x),…,fn(x)}\{ f_1(x), f_2(x), \dots, f_n(x) \}{f1(x),f2(x),…,fn(x)} is said to be _____ on an interval III if there exist constants c1,c2,…,cnc_1, c_2, \dots, c_nc1,c2,…,cn, not all zero, such that c1f1(x)+c2f2(x)+⋯+cnfn(x)=0c_1 f_1(x) + c_2 f_2(x) + \dots + c_n f_n(x) = 0c1f1(x)+c2f2 (x)+⋯+cnfn(x)=0, ∀x∈I\forall x \in I∀x∈I:

- ✓ linear dependent
- ✗ linear independent

1. **In a Bernoulli equation identify \(p(x), q(x) \) and \(n \) respectively.**

- Options:

- \(-y, xy^3, 3\)

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- \(y, xy^3, 3\)
- \(-\frac{1}{x}, x^3, 3\)
- \(\frac{1}{x}, x^3, 3\)
- Correct Answer: \(-y, xy^3, 3\)

2. **The differential equation \(y(1 + 2xy)dx + x(1 - 2xy)dy = 0 \) is**
   - Options:
     - None of these
     - Exact
     - Homogeneous
     - Separable
   - Correct Answer: Exact

3. **Which of the following is an example of an ordinary differential equation?**
   - Options:
     - \(\frac{\partial u}{\partial y} = \frac{\partial v}{\partial x}\)
     - \((y - x) dx + 4xdy = 0\)
     - \(ax^2 + bx + c = 0, a \neq 0\)
     - \(\varphi(x) = f(x) + \lambda \int_a^b k(x, t)\varphi(t)dt\)
   - Correct Answer: \((y - x) dx + 4xdy = 0\)

4. **Which of the following function will satisfy the differential equation: \(\frac{dy}{dx} = -\lambda y\)?**
   - Options:
     - \(y = -ce^{-\lambda x}\)
     - \(y = ce^{\lambda x}\)
     - \(y = ce^{-\lambda x}\)
     - \(y = -ce^{\lambda x}\)
   - Correct Answer: \(y = ce^{-\lambda x}\)

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5. **For the nonexact differential equation \(y(xy + 2x^2y^2) dx + x(xy - x^2y^2) dy = 0 \)**

- Options:

  - \(\frac{1}{x^3y^3}\)

  - \(e^{-x^3y^3}\)

  - \(-\frac{1}{x^3y^3}\)

  - \(e^{x^3y^3}\)

- Correct Answer: \(e^{x^3y^3}\)

6. **The integrating factor for the first order linear differential equation: \(\frac{dy}{dx} + \frac{y}{x \ln x} = \frac{x^2}{\ln x}\) is**

- Options:

  - \(e^x\)

  - \(e^z\)

  - \(\ln|\ln x|\)

  - \(\ln x\)

- Correct Answer: \(\ln x\)

7. **If \(x^2y^3dx + x^3y^2dy = 0\) has the equivalent form as \(\frac{d}{dx}\left(\frac{1}{3} x^3y^3\right) = 0\), then its solution is**

- Options:

  - \(x^3 + y^3 = c\)

  - \(\frac{x^3}{y^3} = c\)

  - \(x^3y^3 = c\)

  - \(x^3 - y^3 = c\)

- Correct Answer: \(x^3 + y^3 = c\)

8. **For \(f(x,y) = \frac{2x}{3y} + 7f(tx,ty) = \ldots\).**

- Options:

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- \(tf(x,y)\)
- \(t^3f(x,y)\)
- \(f(x,y)\)
- Correct Answer: \(f(x,y)\)

9. **On comparing the differential equation: \(dy = dx\) with standard form of \(M(x,y)dx + N(x,y)dy = 0\), we obtain**

- Options:
  - \(|M(x, y)| = |N(x, y)|\)
  - \(M(x, y) = N(x, y)\)
  - \(\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}\)
  - \(\frac{\partial M}{\partial x} \neq \frac{\partial N}{\partial y}\)
- Correct Answer: \(\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}\)

10. **If the non-exact differential equation \(M(x,y)dx + N(x,y)dy = 0\) is homogeneous and \(xM(x, y) + yN(x, y) \neq 0\), then the integrating factor is**

- Options:
  - \(\frac{1}{xM - yN}, \ xM - yN \neq 0\)
  - \(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y} \div M\)
  - \(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \div N\)
  - \(\frac{1}{xM + yN}, \ xM + yN \neq 0\)
- Correct Answer: \(\frac{1}{xM - yN}, \ xM - yN \neq 0\)

1. MCQ: Which of the following would be a constant solution of the separable differential equation: \(\frac{dy}{dx} = e^{x+y} \)?

- A) \(y = a \) where \(a \) is any real number
- B) No constant solution exists
- C) \(y = 0 \)

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- D) $y = 1$

- Correct Answer: C) $y = 0$

2. MCQ: The differential equation $\frac{dx}{dy} + \frac{1}{y} = 2 \sin y$ is first order linear in unknown _____.

- A) $\frac{dy}{dx}$

- B) variable x

- C) multi-variables x and y

- D) variable y

- Correct Answer: D) variable y

3. MCQ: The differential equation $\sin(2x - \tan y)dx - x^2 \sec^2 y dy = 0$ is exact because _____.

- A) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = \sec^2 y$

- B) $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y} = -\sec^2 y$

- C) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = -\sec^2 y$

- D) $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y} = \sec^2 y$

- Correct Answer: A) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = \sec^2 y$

4. MCQ: The differential equation $\frac{d^2 y}{dx^2} + 5 \left(\frac{dy}{dx} \right)^3 - 3y = e^{\sin x}$ is an example of _____ differential equation.

- A) partial linear

- B) ordinary non-linear

- C) ordinary linear

- D) partial non-linear

- Correct Answer: B) ordinary non-linear

5. MCQ: The differential equation $(1 + \ln xy)dx + \left(1 + \frac{x}{y} \right) dy = 0$ is exact because _____.

- A) $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y} = \frac{1}{x}$

- B) $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y} = \frac{1}{y}$

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- C) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = \frac{1}{y}$
- D) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = \frac{1}{x}$
- Correct Answer: C) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} = \frac{1}{y}$

1. MCQ: Equivalent form of the exact differential equation: ydx + xdy = 0.
   - a) d(x/y) = 0
   - b) d(xy) = 0
   - c) d(y/x) = 0
   - d) d(x + y) = 0
   - Correct Answer: b) d(xy) = 0

2. MCQ: For f(x,y) = x^2y - xy^2, f(tx,ty) is...
   - a) t^3 f(x, y)
   - b) f(x, y)
   - c) t^2 f(x, y)
   - d) tf(x, y)
   - Correct Answer: c) t^2 f(x, y)

3. MCQ: Which function would satisfy: dy/dx = d^2y/dx^2 = ... = d^n y/dx^n?
   - a) y = Ax^n e^x
   - b) y = Axe^x
   - c) y = Ae^x^2
   - d) y = Ae^x
   - Correct Answer: d) y = Ae^x

4. MCQ: Order of the differential equation: d^2y/dx^2 + 5(dy/dx)^3 - 3y = e^sin x.
   - a) 3
   - b) 0

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- c) 1

- d) 2

- Correct Answer: d) 2

5. MCQ: Integrating factor for a non-exact differential equation ydx - (y - 3x - 3) dy = 0.

- a) -y^2

- b) y^2

- c) 1/y^2

- d) -1/y^2

- Correct Answer: c) 1/y^2

6. MCQ: Constant solution of the separable differential equation: dy/dx = e^x+y?

- a) y = 0

- b) y = 1

- c) No constant solution exists

- d) y = a ∈ ℝ

- Correct Answer: a) y = 0

7. MCQ: Integrating factor for the first-order linear differential equation: (x - 1)^3 dy/dx + 4(x - 1)^2 y = cos x.

- a) e^(1/(x-1))

- b) e^(x-1)

- c) (x - 1)^4

- d) ln(x - 1)

- Correct Answer: b) e^(x-1)

8. MCQ: Integrating factor for the non-exact differential equation M(x,y)dx + N(x,y)dy = 0.

- a) 1/(xM - yN), xM - yN ≠ 0

- b) ∂M/∂y - ∂N/∂x/N

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- c) ∂N/∂x - ∂M/∂y/M

- d) 1/(xM + yN), xM + yN ≠ 0

- Correct Answer: d) 1/(xM + yN), xM + yN ≠ 0

9. MCQ: The differential equation dx/dy + x/y = 2 sin y is first-order linear in unknown ___.

- a) variable x

- b) multi-variables x and y

- c) variable y

- d) dy/dx

- Correct Answer: c) variable y

10. MCQ: General solution of the separable differential equation: dy/(1+y^2) = 2xdx.

- a) y = cot^(-1)(x^2 + c)

- b) y = tan(x^2 + c)

- c) y = tan^(-1)(x^2 + c)

- d) y = cot(x^2 + c)

- Correct Answer: c) y = tan^(-1)(x^2 + c)

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Which of the following is an equivalent form of the exact differential equation: $ydx + xdy = 0$?

$d(\frac{x}{y}) = 0$

$d(xy) = 0$

$d\left(\frac{y}{x}\right) = 0$

$d(x + y) = 0$

$For f(x, y) = x^2y - xy^2, f(tx, ty) = \ldots.$

$t^3 f(x, y)$

$f(x, y)$

$t^2 f(x, y)$

$tf(x, y)$

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Which of the following function would satisfy: $\frac{dy}{dx} = \frac{d^2y}{dx^2} = \cdots = \frac{d^ny}{dx^n}$?

Select the correct option

- $y = Ax^n e^{x^n}$
- $y = Axe^x$
- $y = Ae^{x^n}$
- $y = Ae^x$

Which of the following function would satisfy: $\frac{dy}{dx} = \frac{d^2y}{dx^2} = \cdots = \frac{d^ny}{dx^n}$?

Select the correct option

- $y = Ax^n e^{x^n}$
- $y = Axe^x$
- $y = Ae^{x^n}$
- $y = Ae^x$

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Which of the following function would satisfy: $\frac{dy}{dx} = \frac{d^2y}{dx^2} = \cdots = \frac{d^ny}{dx^n}$?

Select the correct option

- $y = Ax^n e^{x^n}$
- $y = Axe^x$
- $y = Ae^{x^n}$
- $y = Ae^x$

Which of the following function would satisfy: $\frac{dy}{dx} = \frac{d^2y}{dx^2} = \cdots = \frac{d^ny}{dx^n}$?

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- $y = Axe^x$
- $y = Ae^{x^n}$
- $y = Ae^x$

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Which of the following function would satisfy: $\frac{dy}{dx} = \frac{d^2y}{dx^2} = \cdots = \frac{d^ny}{dx^n}$?

Select the correct option

- $y = Ax^n e^{x^n}$
- $y = Axe^x$
- $y = Ae^{x^n}$
- $y = Ae^x$

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If the non - exact differential equation M(x,y)dx + N(x,y)dy = 0, is of the form yf(xy)dx + xg(xy)dy = 0, then the integrating factor is - - - -

- $\frac{1}{xM - yN}, xM - yN \neq 0$
- $\frac{\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}}{N}$
- $\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M}$
- $\frac{1}{xM + yN}, xM + yN \neq 0$



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The differential equation $\frac{dx}{dy}+\frac{1}{y}x=2\sin y$ *is first order linear in unknown* $---$
variable x
multi - variables x and y
variable y
$\frac{dy}{dx}$

General solution of the separable differential equation: $\frac{dy}{1+y^2}=2xdx$ is---.

Select the correct option

○	$y=\cot^{-1}(x^2+c)$
○	$y=\tan(x^2+c)$
○	$y=\tan^{-1}(x^2+c)$
○	$y=\cot(x^2+c)$



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